

Beacon Innovation Hub

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Data Science

Question 1: Technical Capability Validated/Deployed

- **Coursework Component:** I validated the **NumPy array processing framework** from our data science coursework to handle tabular datasets.
- **Application Details:** I applied this by writing Python scripts in Visual Studio Code designed to convert raw **department spreadsheets** into structured NumPy arrays for faster mathematical computations.
- **Validation Activity:** I explicitly tested array initialization and vectorized operations to replace manual, row-by-row spreadsheet calculations.

Question 2: Operational Friction & Bottlenecks

- **Specific Roadblock:** I am blocked by an **IDE extension conflict** and missing environment paths in Visual Studio Code, resulting in Python activation errors and a missing GitHub CLI (CommandNotFoundException).
- **Impact on Framework:** This operational friction stops me from securely authenticating my environment with GitHub, which prevents me from version-controlling my data science scripts or pushing my NumPy progress to the team repository.

Question 3: Concrete Architectural Change/Workflow Automation for Next Week

- **Measurable Task:** I will fix the environment bottleneck by rebuilding the VS Code extension host and setting up proper Git configuration paths.
- **Live-Test Plan:** Once the environment is stable, I will live-test my NumPy automation script on **three sample department spreadsheets** to measure processing time improvements before our next sync.